

# CHALLENGE - 1

1. Launch one aws ec2 and install jenkins.

Steps: To Install & Troubleshoot

To Install Java 21

```
sudo dnf install java-21-amazon-corretto -y
```

To Install Jenkins:

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/rpm/jenkins.repo
```

```
sudo yum upgrade
```

```
# Add required dependencies for the jenkins package
```

```
sudo yum install fontconfig java-21-openjdk
```

```
sudo yum install jenkins
```

Create a Jenkins systemd override

## Start Jenkins

```
sudo systemctl enable Jenkins
```

```
sudo systemctl start Jenkins
```

```
sudo systemctl status jenkins
```

## For Troubleshoot:

```
sudo systemctl edit Jenkins
```

An editor will open. Add exactly this:

[Service]

```
Environment="JAVA_HOME=/usr/lib/jvm/java-21-amazon-corretto.x86_64"
```

```
Environment="JENKINS_JAVA_CMD=/usr/lib/jvm/java-21-amazon-corretto.x86_64/bin/java"
```

→ 

```
sudo systemctl daemon-reload
```

→ 

```
sudo systemctl restart Jenkins
```

→ 

```
sudo systemctl status jenkins --no-pager
```

To Make Java Default: like if you've installed two java versions

```
sudo alternatives --config java
```

to read password

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

# CHALLENGE - 1

Implementation: Successfully Implementation is Completed

Getting Started

## Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Getting Started

## Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

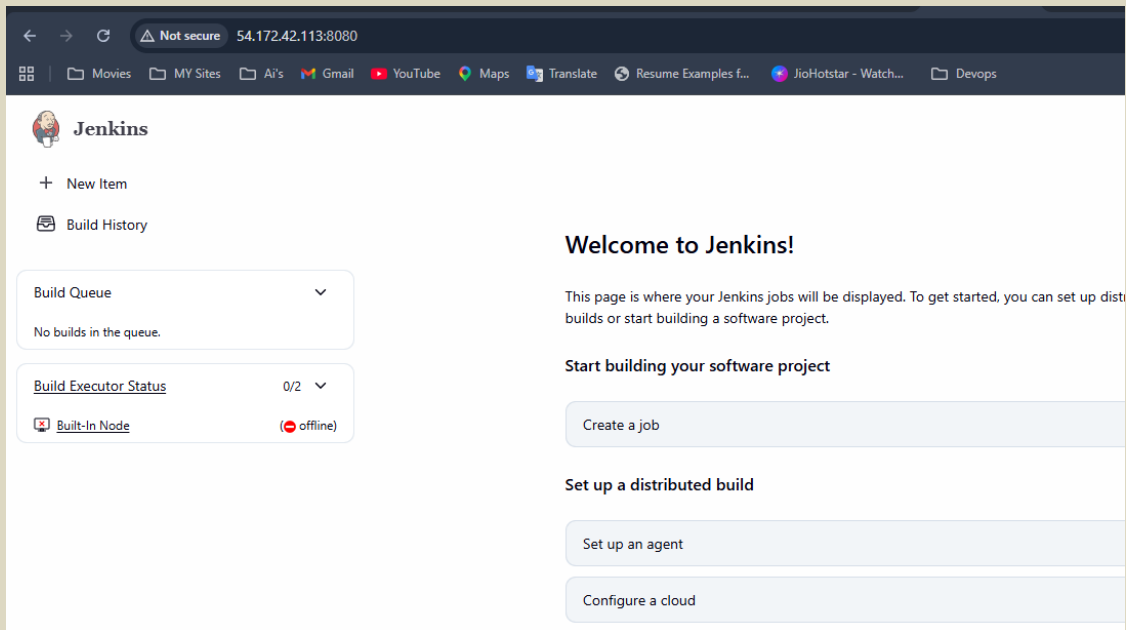
### Install suggested plugins

Install plugins the Jenkins community finds most useful.

### Select plugins to install

Select and install plugins most suitable for your needs.

# CHALLENGE - 1



## 2. Setup Apache Tomcat on Ubuntu Virtual Machine.

```
sudo apt update
```

```
sudo apt upgrade -y
```

```
sudo apt install openjdk-17-jdk -y
```

```
cd /opt/
```

```
wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz
```

```
tar -xvf apache-tomcat-9.0.121.tar.gz
```

```
cd apache-tomcat-9.0.121/
```

```
cd bin/
```

```
./startup.sh
```

# CHALLENGE - 1

```
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sun Aug 30 09:02:22 UTC 2026

System load:  0.62           Temperature:   -273.1 C
Usage of /:   30.6% of 6.61GB Processes:    121
Memory usage: 28%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.27.177

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-27-177:~$
```

Not secure 100.58.122.3:8080

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

## Apache Tomcat/9.0.121

If you're seeing this, you've successfully installed Tomcat. Congratulations!

 Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status  
Manager App  
Host Manager

### Developer Quick Start

- [Tomcat Setup](#)
- [Realms & AAA](#)
- [Examples](#)
- [Servlet Specifications](#)
- [First Web Application](#)
- [JDBC DataSources](#)
- [Tomcat Versions](#)

#### Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the manager

#### Documentation

- [Tomcat 9.0 Documentation](#)
- [Tomcat 9.0 Configuration](#)
- [Tomcat Wiki](#)

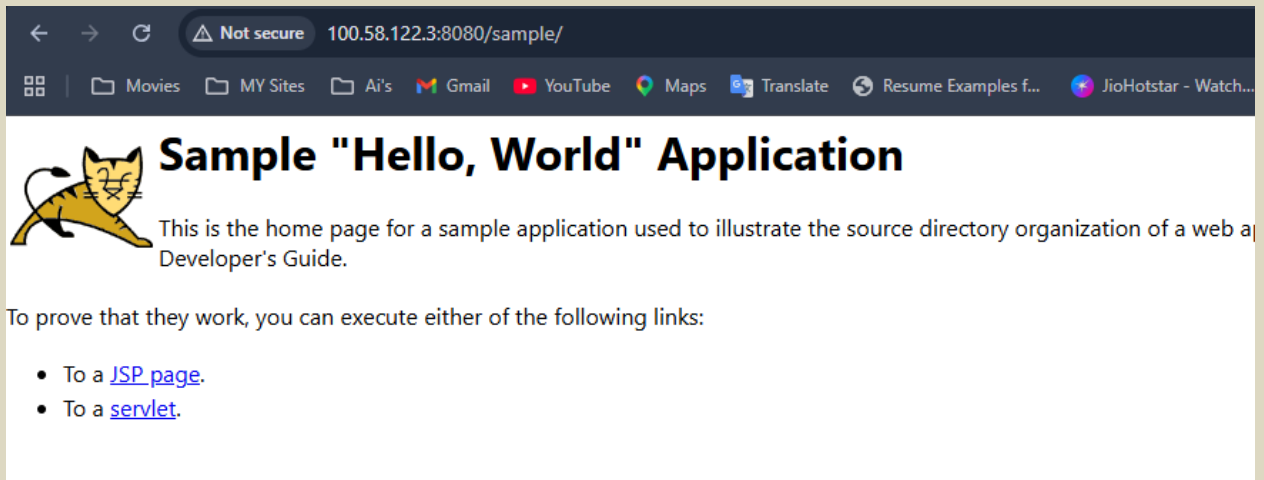
#### Getting Help

##### FAQ and Mailing Lists

The following mailing lists are available:

- [tomcat-announce](#)  
Important announcements, releases, security

# CHALLENGE - 1



3. Setup Docker Desktop on your local windows.

Successfully Docker Desktop Installation Completed

